

The diagram illustrates the wiring for an ESP32-S3 Mini-1-N4R2 module. Key components and connections include:

- Power Supply:** A 3V3 source connected to the module's power pins. A 10K resistor (R4) is connected between 3V3 and GPIO0. A 10uF capacitor (C8) is connected between 3V3 and GND.
- Grounding:** Multiple GND pins are connected to a common ground plane. A 10uF capacitor (C7) is connected between GND and 3V3.
- GPIOs:** Various GPIO pins are connected to different components:
 - GPIO0 to R4
 - GPIO2 to C2
 - GPIO3 to C3
 - GPIO4 to KEY2
 - GPIO5 to KEY3
 - GPIO6 to KEY1
 - GPIO7 to ADC_CTRL1
 - GPIO10 to EPD_BUSY
 - GPIO11 to EPD_RST
 - GPIO12 to EPD_DC
 - GPIO13 to EPD_CS
 - GPIO14 to EPD_CS
 - GPIO15 to RGB
 - GPIO16 to RGB
 - GPIO17 to RGB
 - GPIO18 to RGB
 - GPIO19 to RGB
 - GPIO20 to RGB
 - GPIO21 to RGB
 - GPIO22 to RGB
 - GPIO23 to RGB
 - GPIO24 to RGB
 - GPIO25 to RGB
 - GPIO26 to RGB
 - GPIO27 to RGB
 - GPIO28 to RGB
 - GPIO29 to RGB
 - GPIO30 to RGB
 - GPIO31 to RGB
 - GPIO32 to RGB
 - GPIO33 to RGB
 - GPIO34 to RGB
 - GPIO35 to RGB
 - GPIO36 to RGB
 - GPIO37 to RGB
 - GPIO38 to RGB
 - GPIO39 to RGB
 - GPIO40 to RGB
 - GPIO41 to RGB
 - GPIO42 to RGB
 - GPIO43 to RGB
 - GPIO44 to RGB
 - GPIO45 to RGB
 - GPIO46 to RGB
 - GPIO47 to RGB
 - GPIO48 to RGB
- Other Components:**
 - ADC_CTRL1 connected to ADC_CTRL1
 - EPD_BUSY connected to EPD_BUSY
 - EPD_RST connected to EPD_RST
 - EPD_DC connected to EPD_DC
 - EPD_CS connected to EPD_CS
 - RGB connected to RGB
 - 3V3CTRL connected to 3V3CTRL
 - EPD_DIN connected to EPD_DIN
 - EPD_SCK connected to EPD_SCK
 - nRST connected to nRST
 - RXD connected to RXD
 - TXD connected to TXD
 - SD_CS connected to SD_CS
 - SD_MOSI connected to SD_MOSI
 - SD_CLK connected to SD_CLK
 - SD_MISO connected to SD_MISO
 - SD_D1 connected to SD_D1

按键

[illegible]

调试烧录串口

供电方案	串口信号电压 MCU 工作电压	VDD5 引脚 不低于 V3 电压	V3 引脚 额定 3.3V 左右	V10 引脚 两者用同一电源，1.8V~5V	MCU 或外设电源
全部 USB 供电	5V	USB 供电 5V	仅外接电容	USB 供电 5V	
	3.3V	USB 供电 5V	外接电容	由 V3 供电 3.3V，最多 10mA	
	3.3V	USB 供电 5V 经外置 LDO 电源调节器降压到 3.3V，V3 外接电容			
	1.8V~4V	USB 供电 5V	仅外接电容	USB 供电经外置 LDO 调节器降压	

外设供电开关

PIN	NAME	FUNCTION
1	VOUT	Switch Output.
2	GND	Ground.
3	ILIM	Current Limit Programming Pin. Connect a resistor R_{ILIM} from this pin to GND to program the current limit. $I_{LIM} = \frac{5800}{R_{ILIM}} \text{ (A)}$
4	EN	Chip Enable Pin. Logic high to enable the device. They have integrated a 500kΩ pull-down resistor at this pin.
5	VIN	Switch Input.

烧录USB

电池电量检测

The diagram illustrates a battery level detection circuit. The battery is connected to the BAT_O pin. A 20K resistor (R6) is connected between BAT_O and the gate of the MOSFET (Q3, AO3400A). A 1K resistor (R10) is connected between the gate and the ADC input. A 10K resistor (R13) is connected between the gate and the drain. A 100nF capacitor (C10) is connected between the gate and the drain. The drain is connected to the ADC input. The source is connected to the ADC_CTRL input. A 1K resistor (R16) is connected between the source and the ADC_CTRL input. A 1M resistor (R18) is connected between the source and GND.

LOGO

无线充焊接点

电池充电 电流250ma

$R_{PROG} = 1000V/I_{BAT}$

The diagram shows a battery charging circuit. The input is BAT_I, which goes through a 100nF capacitor (C11) to the GND pin of the TP4054ST25P IC (U5). A 10uF capacitor (C12) is connected between the input and the GND pin. The VCC pin of the IC is connected to a 4.7K resistor (R8) and a CHG LED. The PROG pin of the IC is connected to a 4K resistor (R11) and a 4.7uF/25V capacitor (C13). The output of the IC is VCC.

电源开关 & 电池接口